Formulary for the Machine Learning exam

March 26, 2026

Formulary to group and understand all the topics taught during the machine learning course held at the university "La Sapienza".

1 Introduction and Concept Learning

Target Function Representation

$$\hat{V}(b) = w_0 + w_1 \cdot bp(b) + w_2 \cdot rp(b) + w_3 \cdot bk(b) + w_4 \cdot rk(b) + w_5 \cdot bt(b) + w_6 \cdot rt(b) \quad (1)$$

LMS Weight Update Rule

$$w_i \leftarrow w_i + c \cdot f_i \cdot error(b) \quad (2)$$

Version Space

$$VS_{H,D} \equiv \{h \in H | Consistent(h, D)\} \quad (3)$$

2 Performance Evaluation

True Error

$$error_{\mathcal{D}}(h) \equiv \Pr_{x \in \mathcal{D}}[f(x) \neq h(x)] \quad (4)$$

Sample Error

$$error_S(h) \equiv \frac{1}{n} \sum_{x \in S} \delta(f(x) \neq h(x)) \quad (5)$$

Estimation Bias

$$bias \equiv E_S[error_S(h)] - error_{\mathcal{D}}(h) \quad (6)$$

Confidence Interval

$$error_S(h) \pm z_N \sqrt{\frac{error_S(h)(1 - error_S(h))}{n}} \quad (7)$$

K-Fold Cross Validation Error

$$error_{L,\mathcal{D}} \equiv \frac{1}{k} \sum_{i=1}^{k} \delta_i \quad (8)$$

Comparing Two Hypotheses

$$\hat{d} \equiv error_{S_1}(h_1) - error_{S_2}(h_2) \tag{9}$$

Error Rate

$$\text{Error rate} = \frac{FN + FP}{TP + TN + FP + FN} \tag{10}$$

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

Recall (True Positive Rate)

$$\text{Recall} = \frac{TP}{P} = \frac{TP}{TP + FN} \tag{12}$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP} \tag{13}$$

F1-Score

$$\text{F1-score} = 2\frac{Precision \cdot Recall}{Precision + Recall} \tag{14}$$

Recall, Sensitivity, True Positive Rate (TPR)

$$\text{TPR} = \frac{TP}{P} = \frac{TP}{TP + FN} \tag{15}$$

Specificity, True Negative Rate (TNR)

$$\text{TNR} = \frac{TN}{N} = \frac{TN}{TN + FP} \tag{16}$$

False Positive Rate (FPR)

$$\text{FPR} = \frac{FP}{N} = \frac{FP}{TN + FP} \tag{17}$$

False Negative Rate (FNR)

$$\text{FNR} = \frac{FN}{P} = \frac{FN}{TP + FN} \tag{18}$$

Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\hat{f}(x_i) - t_i| \tag{19}$$

Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n} (\hat{f}(x_i) - t_i)^2 \tag{20}$$

Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{n=1}^{N}(y_n - t_n)^2}{\sum_{n=1}^{N}(t_n - \mu)^2} \tag{21}$$

3 Decision Trees

Entropy (Binary)

$$Entropy(S) \equiv -p_{\oplus} \log_2 p_{\oplus} - p_{\ominus} \log_2 p_{\ominus} \tag{22}$$

Entropy (Multi-class)

$$Entropy(S) \equiv \sum_{i=1}^{c} -p_i \log_2 p_i \tag{23}$$

Information Gain

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) \tag{24}$$

Split Information

$$SplitInformation(S, A) \equiv -\sum_{i=1}^{c} \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|} \tag{25}$$

Gain Ratio

$$GainRatio(S, A) \equiv \frac{Gain(S, A)}{SplitInformation(S, A)} \tag{26}$$

Cost-penalized Gain (Tan and Schlimmer)

$$\frac{Gain^2(S, A)}{Cost(A)} \tag{27}$$

Cost-penalized Gain (Nunez)

$$\frac{2^{Gain(S,A)} - 1}{(Cost(A) + 1)^w} \tag{28}$$

4 Probability and Bayes Networks

Product Rule

$$P(A \wedge B) = P(A|B)P(B) = P(B|A)P(A) \tag{29}$$

Sum Rule

$$P(A \vee B) = P(A) + P(B) - P(A \wedge B) \tag{30}$$

Theorem of Total Probability

$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i) \tag{31}$$

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{32}$$

Marginalization

$$P(\phi) = \sum_{\omega:\omega\models\phi} P(\omega) \tag{33}$$

Chain Rule

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i|X_1, \ldots, X_{i-1}) \tag{34}$$

Conditional Independence

$$P(X, Y|Z) = P(X|Z)P(Y|Z) \tag{35}$$

5 Bayesian Learning

MAP Hypothesis

$$h_{MAP} \equiv \arg\max_{h\in H} P(h|D) = \arg\max_{h\in H} \frac{P(D|h)P(h)}{P(D)} \tag{36}$$

Maximum Likelihood Hypothesis

$$h_{ML} = \arg\max_{h\in H} P(D|h) \tag{37}$$

Bayes Optimal Classifier

$$v_{OB} = \arg\max_{v_j\in V} \sum_{h_i\in H} P(v_j|x, h_i)P(h_i|D) \tag{38}$$

Naive Bayes Classifier

$$v_{NB} = \arg\max_{v_j\in V} P(v_j|D) \prod_i P(a_i|v_j, D) \tag{39}$$

Naive Bayes Estimation (Laplace Smoothing)

$$\hat{P}(a_i|v_j, D) = \frac{|\{\ldots, a_i, \ldots, v_j\}| + mp}{|\{\ldots, v_j\}| + m} \tag{40}$$

Text Classification Naive Bayes

$$c_{NB} = \arg\max_{c_j\in C} P(c_j|D)P(doc|c_j, D) = \arg\max_{c_j\in C} P(c_j) \prod_{i=1}^{\text{length}(doc)} P(w_i|c_j) \tag{41}$$

Bernoulli Distribution

$$P(X = k; \theta) = \theta^k (1-\theta)^{1-k} \tag{42}$$

Multivariate Bernoulli Distribution

$$P(X_1 = k_1, \ldots; \theta_1, \ldots, \theta_n) = \prod_{i=1}^{n} P(X_i = k_i; \theta_i) = \prod_{i=1}^{n} \theta_i^{k_i}(1-\theta_i)^{1-k_i} \tag{43}$$

Binomial Distribution

$$P(X = k; n, \theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k} \tag{44}$$

Multinomial Distribution

$$P(X_1 = k_1, \ldots, X_d = k_d; n, \theta_1, \ldots, \theta_d) = \frac{n!}{k_1! \ldots k_d!}\theta_1^{k_1} \ldots \theta_d^{k_d} \tag{45}$$

General Maximum Likelihood Estimation

$$\Theta_{ML} = \arg\max_{\Theta} \log P(D|\Theta) \tag{46}$$

6 Probabilistic Models for Classification

Generative Model Posterior (2 classes)

$$P(C_1|\mathbf{x}) = \sigma(a) = \frac{1}{1+\exp(-a)} \tag{47}$$

Linear Argument (a)

$$a = \ln \frac{P(\mathbf{x}|C_1)P(C_1)}{P(\mathbf{x}|C_2)P(C_2)} = \mathbf{w}^T\mathbf{x} + w_0 \tag{48}$$

Gaussian Linear Parameters

$$\begin{aligned} \mathbf{w} &= \Sigma^{-1}(\mu_1 - \mu_2) \\ w_0 &= -\frac{1}{2}\mu_1^T\Sigma^{-1}\mu_1 + \frac{1}{2}\mu_2^T\Sigma^{-1}\mu_2 + \ln\frac{\pi}{1-\pi} \end{aligned} \tag{49}$$

Maximum Likelihood for Gaussian Parameters

$$\begin{aligned} \hat{\pi} &= \frac{N_1}{N} \\ \hat{\mu}_1 &= \frac{1}{N_1}\sum_{n=1}^{N} t_n\mathbf{x}_n, \quad \hat{\mu}_2 = \frac{1}{N_2}\sum_{n=1}^{N}(1-t_n)\mathbf{x}_n \\ \hat{\Sigma} &= \frac{N_1}{N}S_1 + \frac{N_2}{N}S_2 \end{aligned} \tag{50}$$

Generative Model Posterior (K classes)

$$P(C_k|\mathbf{x}) = \frac{\exp(a_k)}{\sum_j \exp(a_j)} \tag{51}$$

Logistic Regression Cross-Entropy

$$E(\tilde{\mathbf{w}}) \equiv -\ln p(\mathbf{t}|\tilde{\mathbf{w}}) = -\sum_{n=1}^{N}[t_n \ln y_n + (1-t_n)\ln(1-y_n)] \tag{52}$$

Logistic Regression Gradient

$$\nabla E(\tilde{\mathbf{w}}) = \sum_{n=1}^{N}(y_n - t_n)\tilde{\mathbf{x}}_n \tag{53}$$

Iterative Reweighted Least Squares (IRLS) Update

$$\begin{aligned} \tilde{\mathbf{w}} &\leftarrow \tilde{\mathbf{w}} - \mathbf{H}(\tilde{\mathbf{w}})^{-1}\nabla E(\tilde{\mathbf{w}}) \\ \mathbf{H}(\tilde{\mathbf{w}}) = \nabla\nabla E(\tilde{\mathbf{w}}) &= \sum_{n=1}^{N} y_n(1-y_n)\tilde{\mathbf{x}}_n\tilde{\mathbf{x}}_n^T = \tilde{\mathbf{X}}^T\mathbf{R}\tilde{\mathbf{X}} \end{aligned} \tag{54}$$

Multiclass Logistic Regression Cross-Entropy

$$E(\tilde{\mathbf{w}}_1, \ldots, \tilde{\mathbf{w}}_K) = -\sum_{n=1}^{N}\sum_{k=1}^{K} t_{nk}\ln y_{nk} \tag{55}$$

7 Linear Models for Classification

Linear Discriminant Function

$$y(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + w_0 = \tilde{\mathbf{w}}^T\tilde{\mathbf{x}} \tag{56}$$

Least Squares Error (Matrix Form)

$$E(\mathbf{W}) = \frac{1}{2}\mathrm{Tr}\left\{(\mathbf{XW}-\mathbf{T})^T(\mathbf{XW}-\mathbf{T})\right\} \tag{57}$$

Least Squares Closed-Form Solution

$$\mathbf{W}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{T} = \mathbf{X}^\dagger\mathbf{T} \tag{58}$$

Perceptron Output

$$o(\mathbf{x}) = \mathrm{sign}(\mathbf{w}^T\mathbf{x}) \tag{59}$$

Perceptron Squared Error

$$E(\mathbf{w}) \equiv \frac{1}{2}\sum_{n=1}^{N}(t_n - o_n)^2 = \frac{1}{2}\sum_{n=1}^{N}(t_n - \mathbf{w}^T\mathbf{x}_n)^2 \tag{60}$$

Perceptron Weight Update

$$\Delta w_i = \eta\sum_{n=1}^{N}(t_n - \mathrm{sign}(\mathbf{w}^T\mathbf{x}_n))x_{i,n} \tag{61}$$

Fisher's Criterion

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}} \tag{62}$$

Fisher Linear Discriminant Solution

$$\mathbf{w}^* \propto \mathbf{S}_W^{-1}(\mathbf{m}_2 - \mathbf{m}_1) \tag{63}$$

Support Vector Machine Canonical Constraint

$$\mathbf{w}^*, w_0^* = \arg\min \frac{1}{2}||\mathbf{w}||^2 \quad \text{subject to} \quad t_n(\mathbf{w}^T \mathbf{x}_n + w_0) \geq 1 \quad \forall n \tag{64}$$

Soft Margin SVM Objective

$$\mathbf{w}^*, w_0^* = \arg\min \frac{1}{2}||\mathbf{w}||^2 + C\sum_{n=1}^{N} \xi_n \quad \text{subject to} \quad t_n(\mathbf{w}^T \mathbf{x}_n + w_0) \geq 1 - \xi_n \tag{65}$$

SVM Lagrangian Optimization Problem

$$\tilde{L}(\mathbf{a}) = \sum_{n=1}^{N} a_n - \frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N} a_n a_m t_n t_m \mathbf{x}_n^T \mathbf{x}_m \tag{66}$$

SVM Hyperplane Support Vectors

$$\mathbf{w}^* = \sum_{n=1}^{N} a_n^* t_n \mathbf{x}_n \tag{67}$$

8 Linear Models for Regression

Linear Regression Basis Function Model

$$y(\mathbf{x}; \mathbf{w}) = \sum_{j=0}^{M} w_j \phi_j(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) \tag{68}$$

Linear Regression Log-Likelihood

$$\ln P(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta) = -\beta \frac{1}{2}\sum_{n=1}^{N}[t_n - \mathbf{w}^T \phi(\mathbf{x}_n)]^2 - \frac{N}{2}\ln(2\pi\beta^{-1}) \tag{69}$$

Linear Regression Least Squares Solution

$$\mathbf{w}_{ML} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{t} = \Phi^\dagger \mathbf{t} \tag{70}$$

SGD Update

$$\hat{\mathbf{w}} \leftarrow \hat{\mathbf{w}} - \eta \nabla E_S \tag{71}$$

Linear Regression Regularized Error

$$\arg\min_{\mathbf{w}} E_D(\mathbf{w}) + \lambda E_W(\mathbf{w}) \tag{72}$$

9 Kernel Methods

Kernelized Linear Model Dual Solution

$$\alpha = (\mathbf{K} + \lambda \mathbf{I}_N)^{-1}\mathbf{t} \quad \text{where } \mathbf{K} = \mathbf{X}\mathbf{X}^T \tag{73}$$

Polynomial Kernel

$$k(\mathbf{x}, \mathbf{x}') = (\beta \mathbf{x}^T \mathbf{x}' + \gamma)^d \tag{74}$$

Radial Basis Function (RBF) Kernel

$$k(\mathbf{x}, \mathbf{x}') = \exp(-\beta ||\mathbf{x} - \mathbf{x}'||^2) \tag{75}$$

Sigmoid Kernel

$$k(\mathbf{x}, \mathbf{x}') = \tanh(\beta \mathbf{x}^T \mathbf{x}' + \gamma) \tag{76}$$

Kernelized SVM Regression Loss

$$E(\mathbf{w}) = C \sum_{n=1}^{N} E_\epsilon(y_n, t_n) + \frac{1}{2}||\mathbf{w}||^2 \tag{77}$$

Kernelized SVM Prediction

$$y(\mathbf{x}) = \sum_{n=1}^{N} (\hat{a}_n - \hat{a}'_n) k(\mathbf{x}, \mathbf{x}_n) + \hat{w}_0 \tag{78}$$

10 Instance Based Learning

K-Nearest Neighbors Likelihood

$$p(c|\mathbf{x}, D, K) = \frac{1}{K} \sum_{\mathbf{x}_n \in N_K(\mathbf{x}, D)} \mathbb{I}(t_n = c) \tag{79}$$

Kernelized Distance

$$||\mathbf{x} - \mathbf{x}_n||^2 = \mathbf{x}^T\mathbf{x} + \mathbf{x}_n^T\mathbf{x}_n - 2\mathbf{x}^T\mathbf{x}_n \tag{80}$$

11 Artificial Neural Networks

Network Layer Activation

$$\mathbf{h}_{i+1} = f(\mathbf{W}_i^T \mathbf{h}_i + \mathbf{b}_i) \tag{81}$$

Cross-Entropy Loss

$$J(\theta) = \mathbb{E}_{\mathbf{x},\mathbf{t}\sim\mathcal{D}}[-\ln(p(\mathbf{t}|\mathbf{x}, \theta))] \tag{82}$$

Softmax Activation

$$y_i = \text{softmax}(\alpha^{(i)}) = \frac{\exp(\alpha^{(i)})}{\sum_j \exp(\alpha_j)} \tag{83}$$

Rectified Linear Unit (ReLU)

$$g(\alpha) = \max(0, \alpha) \tag{84}$$

Sigmoid Activation

$$g(\alpha) = \sigma(\alpha) = \frac{1}{1 + \exp(-\alpha)} \tag{85}$$

Hyperbolic Tangent Activation

$$g(\alpha) = \tanh(\alpha) = 2\sigma(2\alpha) - 1 \tag{86}$$

Chain Rule

$$\nabla_{\mathbf{x}} z = \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}}\right)^T \nabla_{\mathbf{y}} z \tag{87}$$

SGD with Momentum Update

$$\begin{aligned} \mathbf{v}^{(k+1)} &\leftarrow \mu \mathbf{v}^{(k)} - \eta \mathbf{g} \\ \theta^{(k+1)} &\leftarrow \theta^{(k)} + \mathbf{v}^{(k+1)} \end{aligned} \tag{88}$$

Parameter Norm Penalty

$$\bar{J}(\theta) = J(\theta) + \lambda E_{reg}(\theta) \quad \text{with} \quad E_{reg}(\theta) = \sum_j |\theta_j|^q \tag{89}$$

12 Convolutional Neural Networks

Continuous Convolution

$$(x * w)(t) \equiv \int_{-\infty}^{\infty} x(a) w(t-a) da \tag{90}$$

Discrete 2D Convolution

$$(I * K)(i, j) \equiv \sum_{m \in S_1} \sum_{n \in S_2} I(m, n) K(i - m, j - n) \tag{91}$$

Discrete 2D Cross-Correlation

$$(I * K)(i, j) = \sum_{m \in S_1} \sum_{n \in S_2} I(i + m, j + n) K(m, n) \tag{92}$$

Feature Map Dimensions

$$w_{out} = \frac{w_{in} - w_k + 2p}{s} + 1, \quad h_{out} = \frac{h_{in} - h_k + 2p}{s} + 1 \tag{93}$$

Convolutional Layer Parameter Count

$$|\theta| = w_k \cdot h_k \cdot d_{in} \cdot d_{out} + d_{out} \tag{94}$$

13 Multiple Learners

Voting (Regression)

$$y_{voting}(\mathbf{x}) = \sum_{m=1}^{M} w_m y_m(\mathbf{x}) \tag{95}$$

Voting (Classification)

$$y_{voting}(\mathbf{x}) = \arg\max_c \sum_{m=1}^{M} w_m \mathbb{I}(y_m(\mathbf{x}) = c) \tag{96}$$

Bagging Prediction

$$y_{bagging}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^{M} y_m(\mathbf{x}) \tag{97}$$

AdaBoost Final Classifier

$$Y_M(\mathbf{x}) = \text{sign}\left(\sum_{m=1}^{M} \alpha_m y_m(\mathbf{x})\right) \tag{98}$$

AdaBoost Weighted Error

$$\epsilon_m = \frac{\sum_{n=1}^{N} w_n^{(m)} \mathbb{I}(y_m(\mathbf{x}_n) \neq t_n)}{\sum_{n=1}^{N} w_n^{(m)}} \tag{99}$$

AdaBoost Alpha Parameter

$$\alpha_m = \ln\left(\frac{1 - \epsilon_m}{\epsilon_m}\right) \tag{100}$$

AdaBoost Weight Update

$$w_n^{(m+1)} = w_n^{(m)} \exp[\alpha_m \mathbb{I}(y_m(\mathbf{x}_n) \neq t_n)] \tag{101}$$

Bias-Variance Decomposition

$$\begin{aligned} MSE &= E_S[(y - \hat{f}(\mathbf{x}; D))^2] \\ &= Bias^2 + Variance + Noise \\ Bias &= E_S[\hat{f}(\mathbf{x}; D)] - f(\mathbf{x}) \\ Variance &= E_S[(\hat{f}(\mathbf{x}; D) - E_S[\hat{f}(\mathbf{x}; D)])^2] \end{aligned} \tag{102}$$

14 Unsupervised Learning

Gaussian Mixture Model

$$P(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}; \mu_k, \Sigma_k) \tag{103}$$

GMM Posterior Responsibilities (E-step)

$$\gamma(z_k) \equiv P(z_k = 1|\mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x}; \mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}; \mu_j, \Sigma_j)} \tag{104}$$

GMM Parameters Update (M-step)

$$\begin{aligned} \mu_k &= \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) \mathbf{x}_n \\ \Sigma_k &= \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk})(\mathbf{x}_n - \mu_k)(\mathbf{x}_n - \mu_k)^T \\ \pi_k &= \frac{N_k}{N}, \quad \text{with} \quad N_k = \sum_{n=1}^{N} \gamma(z_{nk}) \end{aligned} \tag{105}$$

General EM Algorithm

$$\begin{aligned} &\text{E-step: } Q(\theta'|\theta) \leftarrow \mathbb{E}[\ln P(\mathbf{Y}|\theta')|\theta, \mathbf{X}] \\ &\text{M-step: } \theta \leftarrow \arg\max_{\theta'} Q(\theta'|\theta) \end{aligned} \tag{106}$$

15 Dimensionality Reduction

PCA Covariance Matrix

$$S = \frac{1}{N} \sum_{n=1}^{N} (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T = \frac{1}{N} \mathbf{X}^T \mathbf{X} \tag{107}$$

PCA Projected Variance Maximization

$$\max_{\mathbf{u}_1} \mathbf{u}_1^T \mathbf{S} \mathbf{u}_1 \quad \text{subject to} \quad \mathbf{u}_1^T \mathbf{u}_1 = 1 \tag{108}$$

PCA Approximation Error

$$J = \frac{1}{N} \sum_{n=1}^{N} ||\mathbf{x}_n - \tilde{\mathbf{x}}_n||^2 = \sum_{i=m+1}^{d} \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i \tag{109}$$

PCA High-Dimensional Eigenvector

$$\frac{1}{N} \mathbf{X}\mathbf{X}^T \mathbf{v}_i = \lambda_i \mathbf{v}_i \implies \mathbf{u}_i = \frac{1}{\sqrt{N\lambda_i}} \mathbf{X}^T \mathbf{v}_i \tag{110}$$

Probabilistic PCA Model

$$P(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; \mathbf{W}\mathbf{z} + \mu, \sigma^2 \mathbf{I}) \tag{111}$$

Probabilistic PCA Marginal Distribution

$$P(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mu, \mathbf{C}) \quad \text{with} \quad \mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2 \mathbf{I} \tag{112}$$

Probabilistic PCA Posterior Distribution

$$P(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}; \mathbf{M}^{-1}\mathbf{W}^T(\mathbf{x} - \mu), \sigma^2 \mathbf{M}) \quad \text{with} \quad \mathbf{M} = \mathbf{W}^T\mathbf{W} + \sigma^2 \mathbf{I} \tag{113}$$

16 Markov Decision Processes and Reinforcement Learning

HMM Joint Probability

$$P(x_{0:T}, z_{1:T}) = P(x_0)P(z_0|x_0)P(x_1|x_0)P(z_1|x_1)\dots \tag{114}$$

HMM Forward Step

$$\alpha_t^k \equiv P(x_t = k, z_{1:t}) = b_k(z_t)\sum_j \alpha_{t-1}^j A_{jk} \tag{115}$$

HMM Backward Step

$$\beta_t^k \equiv P(z_{t+1:T}|x_t = k) = \sum_j \beta_{t+1}^j A_{kj} b_j(z_{t+1}) \tag{116}$$

HMM Smoothing

$$P(x_t = k|z_{1:T}) = \frac{\alpha_t^k \beta_t^k}{\sum_j \alpha_t^j \beta_t^j} \tag{117}$$

POMDP Belief State Update

$$b'(x') = \frac{o(x', a, z')\sum_{x\in X}\delta(x, a, x')b(x)}{P(z'|b, a)} \tag{118}$$

POMDP Value Function

$$V(b) = \max_{a\in A}\left[\sum_{x\in X} b(x)r(x, a) + \gamma\sum_{z\in Z} P(z|b, a)V(b_z^a)\right] \tag{119}$$

MDP Value Function (Stochastic)

$$V^\pi(x) \equiv \mathbb{E}[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots] \tag{120}$$

MDP Bellman Equation

$$V^\pi(x) = \sum_{x'}\delta(x'|x, \pi(x))\left(r(x, \pi(x), x') + \gamma V^\pi(x')\right) \tag{121}$$

Q Function (Deterministic)

$$Q^\pi(x, a) \equiv r(x, a) + \gamma V^\pi(\delta(x, a)) \tag{122}$$

Q Function (Stochastic)

$$Q(x, a) \equiv \mathbb{E}[r(x, a)] + \gamma\sum_{x'} P(x'|x, a)\max_{a'} Q(x', a') \tag{123}$$

Optimal Policy from Q

$$\pi^*(x) = \arg\max_{a\in A} Q(x, a) \tag{124}$$

Q-learning Update (Deterministic)

$$\hat{Q}_{(t)}(x,a) \leftarrow \bar{r} + \gamma \max_{a' \in A} \hat{Q}_{(t-1)}(x',a') \tag{125}$$

Q-learning Update (Non-Deterministic)

$$\hat{Q}_n(x,a) \leftarrow (1-\alpha)\hat{Q}_{n-1}(x,a) + \alpha \left[r + \gamma \max_{a'} \hat{Q}_{n-1}(x',a') \right] \tag{126}$$

SARSA Update

$$\hat{Q}_n(x,a) \leftarrow \hat{Q}_{n-1}(x,a) + \alpha \left[r + \gamma \hat{Q}_{n-1}(x',a') - \hat{Q}_{n-1}(x,a) \right] \tag{127}$$

Soft-max Exploration Strategy

$$P(a_i|x) = \frac{k^{\hat{Q}(x,a_i)}}{\sum_j k^{\hat{Q}(x,a_j)}} \tag{128}$$

Policy Gradient Update

$$\theta \leftarrow \theta + \eta \Delta_\theta \rho(\theta) \tag{129}$$